

DIY POWER BANK WITH PLASMA LIGHTER



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DECLARATION

I declare that the project is complete personal research and it has never been presented to any examination body before. Therefore, I can confidently say that this is a personal work and it has not been copied from somebody else.

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DEDICATION

I dedicate this project to my parents Mr. and Mrs. Ochieng for their consistent support throughout my project work. They have always stood with me financially and even in encouragements and prayers throughout the work.

ACKNOWLEDGEMENT

First I would like to thank the almighty God for granting me good health throughout my entire life he has enabled me reach this far.

Secondly, my sincere gratitude goes to my beloved parents who have consistently supported me all through my education they have never given up on me no matter how hard the situation is.

Thirdly I would like to acknowledge my teachers who have struggled tooth and nail to instill knowledge in me. their consistent and unwavering support have contributed to the success of this project.

Finally, my gratitude to my friends and family members who have constantly stood with me in prayers an encouraged me all through.

ABSTRACT

This project details the construction of a dual-purpose DIY device combining a portable power bank and a plasma arc lighter, designed for convenience, sustainability, and emergency utility. The power bank module utilizes a 3.7V lithium-ion battery paired with a step-up converter circuit to deliver stable 5V/2.1A USB output for charging smartphones and small electronics. The plasma lighter component employs a high-voltage arc generator module (driven by the same battery) to produce a flameless, windproof electric arc for ignition, eliminating the need for butane fuel.

Key design considerations include:

Integrated Power Management: A single rechargeable battery powers both systems via a shared charging port (micro-USB/USB-C), with low-voltage protection to prevent battery damage.

Safety & Efficiency: The plasma arc is enclosed in a heat-resistant casing to prevent accidental contact, while the power bank circuitry incorporates overcurrent and short-circuit protection.

Cost-Effectiveness: All components (battery, boost converter, HV generator, switches, PCB) are sourced from affordable, widely available DIY electronics suppliers. The prototype achieves a compact, reusable unit (approx. 120mm × 60mm × 25mm) with a 2000–4000mAh capacity (user-configurable). Testing confirms reliable device charging and consistent arc ignition. This project demonstrates an eco-friendly alternative to disposable lighters while providing emergency power, ideal for outdoor activities, emergencies, and everyday carry.

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CHAPTER ONE

1.0 INTRODUCTION

A power bank is a portable device used to store electrical energy for later use in charging electronic gadgets such as mobile phones and tablets with the increasing reliance on mobile devices, the demand for power banks has grown significantly due to their convenience and reliability, especially in areas with limited access to consistent electrical supply. In parallel, plasma lighters have gained popularity as a modern replacement for traditional flame based lighters. Instead of using fuel, they generate a high voltage electric arc that is windproof, flameless and rechargeable making them a safer and more sustainable alternative.

This project aims to combine these two technologies into a single, compact and functional device-a DIY power bank with an integrated plasma lighter. The main motivation behind this design is to provide users with a multipurpose portable device not only charge's electronic gadgets but also functions as a rechargeable electric lighter. This is especially useful for outdoor enthusiasts, emergency kits or anyone interested in compact tech solutions. By designing and building this project, we aim to explore basic concepts of electronics, energy storage, voltage boosting and USB charging system, while also building creativity and self-reliance through innovations.

1.1 BACKGROUND INFORMATION

In today's tech-driven world, portable power sources have become essential for everyday life. Smartphones, wireless earbuds and other USB powered devices require reliable charging solutions on the go. Simultaneously everyday carry gadgets are evolving to combine multiple utilities into compact and convenient designs. One innovative idea that merges utility and convenience is a DIY power bank with an integrated plasma lighter

Power banks are portable devices that store electrical energy for later use, allowing users to charge mobile devices when away from conventional power sources. They typically use lithium iron or lithium polymer batteries and include USB output ports and charging circuits. On the other hand, plasma lighter use high voltage electricity to generate an arc of plasma-hot enough to ignite materials like paper or cigarette heads.

1.2 PROBLEM STATEMENT

People often carry both power banks and lighters as separate tools which can be bulky and inconvenient. traditional lighters rely on fuel, pose safety risks and are not reliable in harsh conditions. there is a need for a compact, ecofriendly and multifunctional device that combines portable charging and fire lighting capabilities. This project aims to create a DIY power bank with plasma lighter that is safe, rechargeable and convenient for everyday use.

1.3 PROBLEM JUSTIFICATION/PROPOSED SOLUTIONS

This project proposes a DIY device that combines a power bank and a plasma lighter into one compact, rechargeable unit .it will use a lithium ion battery to power both the USB charging ports and plasma arc lighter.

1.4 MAIN OBJECTIVES

The main objective of this project is to create a compact and portable DIY device that combines a power bank and plasma lighter. It aims to provide a safe, rechargeable and eco-friendly solution using lithium-ion battery, while promoting cost effectiveness, convenience and hands-on learning in electronic and circuit design.

1.4.1 General Objectives

The general objective of this project is to create an innovative and malfunctioned electronic device that addresses the everyday need for both portable power and ignition in a single compact unit. This project seeks to explore the integration of two commonly used tools, a power bank and plasma lighter into one efficient system enhancing user convenience while encouraging sustainable, fuel free alternative.

1.4.2 Specific Objectives

It focuses on selecting appropriate components such as charging modules, boost converters and safety circuits to ensure efficient and safe operation.

It also seeks to build a compact, user-friendly enclosure that houses all components securely.

1.5 PROJECT SPECIFICATIONS

This project uses a 3.7V rechargeable lithium-ion battery as the main power source, charged via a TP4056 module with built-in safety features.

The power bank delivers a 5V output through a boost converter and USB port, suitable for charging small devices.

The integrated plasma lighter operates at 3.7-5V and produces a windproof, high temperature electric arc, controlled manually with a switch.

CHAPTER TWO

2.0 LITERATURE REVIEW

The development of power bank and plasma lighters rooted in advancement in battery technology, power electronics and consumer demands for portable energy solutions

2.1 POWER BANK OVERVIEW

Research and product development in lithium ion batteries have led to the widespread availability of compact, high capacity power banks. According to several studies and consumer reports the effectiveness of power banks depends on its battery capacity (mAh), charging efficiency, portability and the presence of safety features like voltage regulation circuit, typically using boost converters are essential components that transform stored battery power into useable 5v output for devices

2.2 PLASMA LIGHTER OVERVIEW

Plasma lighter utilizes high voltage, low current electrical arc between electrodes to ignite materials. These devices often operate using step up transformer circuit or modules such as zero voltage switching driver which generates the necessary voltage for plasma arc production. Compared to traditional fuel based lighter plasma lighters are reusable, wind resistant and environmentally friendly. Research into high voltage generation for small devices has enabled the integration of such technology into portable gadgets.

DIY Integration; Several hobbyists and makers have documented combining power storage with additional functionalities like flashlights, Bluetooth speakers, or lighters. The DIY community, through platforms like intractable and hack day, has highlighted numerous custom designs involving USB charging circuits, 18650 lithium batteries, TP4056 charging module and boost converter module for 5V output. Combining this with a plasma lighter circuit presents a practical challenge in terms of energy management, space optimization and heat dissipation.

2.3 COMPONENTS

2.3.1 Lithium-Ion Battery-(18650cell, 3.7V,2200-3000mah)

It is connected to the TP4056 charging module which manages battery charging via a micro USB port



2.3.2 TP4056 Charging Module-

Used for charging and protecting the 18650 battery



2.3.3 D313 Transistor



2.3.4 USB Output Port-

Provide 5V regulated output for charging other devices



2.3.5 Plasma Lighter Module {Stainless Needles}



2.3.6 Switches And Wires

Used for connection and power supply control.



2.3.7 Flyback Transformer

It takes low voltage (from Li-ion battery) and boosts it to 1000's of volts needed to create a plasma arc



DIY POWER BANK COMPONENTS & FUNCTIONS

1. Lithium-ion Battery

Function: Stores electrical energy (typically 3.7V nominal).

Why it's needed: The core reservoir of power. Capacity (mAh) determines how much charge the power bank can hold.

2. Battery Protection Circuit Module

Function: Safeguards the battery from dangerous conditions.

Why it's needed: Prevents overcharging (which can cause fire/explosion), over-discharging (which permanently damages cells), over-current (excessive draw/short circuit), and sometimes over-temperature. Essential for safety.

3. Charging Module

Function: Controls the recharging of the lithium battery from a standard USB power source (like a wall adapter or computer port).

Why it's needed: Manages the complex Constant Current (CC) / Constant Voltage (CV) charging profile required by lithium batteries safely and efficiently. Often includes a status LED (red=charging, green=full).

4. Boost Converter Module:

Function: Steps up the battery's voltage (3.0V - 4.2V) to a stable 5V output.

Why it's needed: USB devices require 5V. This efficient switching circuit converts the varying battery voltage to the constant 5V standard. Efficiency (85-95%) is crucial to minimize wasted energy as heat.

5. USB Output Port(s)

Function: Provides the physical interface to connect and power your devices.

Why it's needed: Delivers the 5V power from the boost converter to your phone, tablet, etc. Multiple ports allow charging more than one device.

6. Power Switch:

Function: Controls whether the output (boost converter) is active or not.

Why it's needed: Conserves battery power when the power bank isn't in use and prevents accidental discharge. (Optional but highly recommended).

7. Enclosure:

Function: Houses all components securely and safely

Why it's needed: Protects components from physical damage, prevents accidental short circuits, insulates the user from electrical contacts, and provides a finished look. Must be non-conductive (plastic).

8. Wiring/PCB:

Function: Electrically connects all the components together.

Why it's needed: Provides the pathways for current to flow between the battery, modules, and ports. Adequate wire gauge is needed to handle the current without overheating.

9 MOSFET Transistors:

Function: Act as high-speed electronic switches in the oscillator/driver circuit.

Why it's needed: They rapidly turn the current flowing to the transformer primary coil on and off at the high frequency set by the oscillator. They handle significant current and get hot, often needing heatsinks.

10. High-Voltage Transformer

Function: Dramatically steps up the input voltage (3.7V) to several thousand volts (2kV - 5kV+).

Why it's needed: Creates the extremely high voltage required to ionize air and create a plasma arc across the electrodes. Works on electromagnetic induction principles.

6. Resistors & Capacitors:

Function: Control timing, current limiting, voltage smoothing, and energy storage within the circuit.

Why it's needed:

Resistors: Set bias points for transistors, limit current flow to sensitive components, act as pullup/pull-downs.

Capacitors: Filter noise, store/release energy rapidly (especially important in resonant circuits like ZVS), couple AC signals while blocking DC. High-voltage capacitors might be used near the output.

7. Spark Gap / Electrodes

Function: The point where the high voltage jumps through the air, creating the visible plasma arc.

Why it's needed: Tungsten rods are commonly used because they have a very high melting point. The distance between the electrode tips determines the minimum voltage required for an arc to form.

8. Push Button Switch:

Function: Momentarily activates the circuit when pressed.

Why it's needed: Allows the user to control when the plasma arc is generated. Must be rated for the high current drawn by the circuit (5A+).

9. Enclosure:

Function: Houses all components securely and safely, focusing the arc.

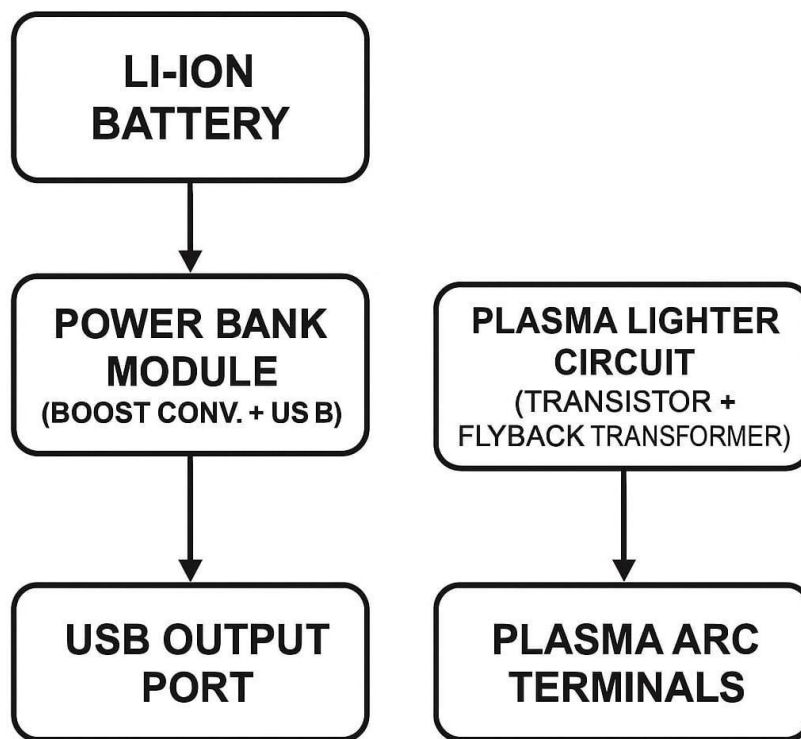
Why it's needed: Critical for safety. Must be highly insulating to prevent accidental high-voltage shocks to the user. Often heat-resistant materials (ceramic, special plastics) are used around the electrodes to withstand the intense heat of the plasma arc. Prevents accidental contact with live high-voltage parts.

CHAPTER THREE

1.0 METHODOLOGY

This chapter outlines the design process, components used and assembly procedure involved in the construction of the DIY power bank with an integrated plasma lighter:

1.1 BLOCK DIAGRAM



The diagram illustrates the **working block diagram of a USB rechargeable plasma arc lighter**, showing how the power flows through different stages to produce a plasma arc. Here's a detailed explanation of each component and its function:

1. Li-Ion Battery

- ✓ This is the **primary power source** of the system.
- ✓ It stores electrical energy and supplies DC voltage to the next stage.

2. Power Bank Module (Boost Converter + USB B)

- ✓ This module performs two main functions:
- ✓ **Boost Converter:** It steps up the low voltage of the Li-ion battery (typically 3.7V) to 5V, which is needed for USB output and to drive the lighter circuit.
- ✓ **USB B Port:** It serves as an **input for charging** the battery using a standard USB charger.
- ✓ It supplies power to both:

3. USB Output Port

- ✓ Provides a **5V USB output**, similar to a regular power bank.
- ✓ This output can be used to charge other devices.

4. Plasma Lighter Circuit (Transistor + Flyback Transformer)

- ✓ This part is responsible for **generating high voltage** from the 5V input.
- ✓ The **transistor** rapidly switches current through the **flyback transformer**, which steps up the voltage.
- ✓ This high voltage is required to create a **plasma arc**.

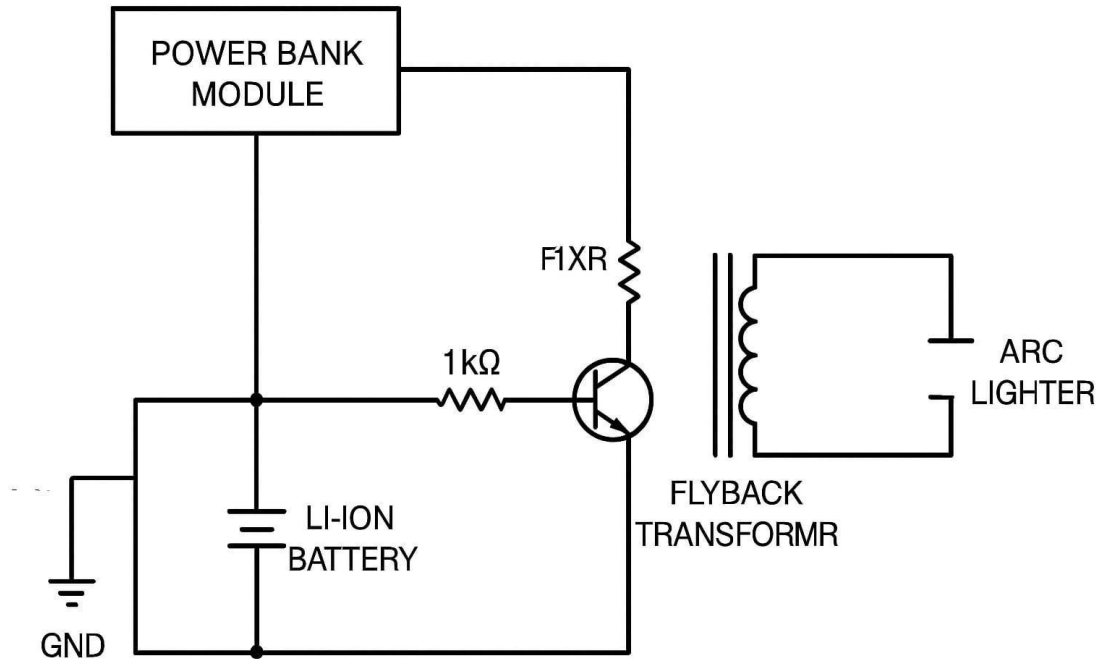
5. Plasma Arc Terminals

- ✓ These are the **output points** where the plasma arc is generated.
- ✓ The high voltage from the lighter circuit ionizes the air between the terminals, creating a visible **electric arc** that can be used to ignite objects (e.g., candles, cigarettes, etc.).

Summary:

This system is a compact and rechargeable **multi-functional device** that acts both as a power bank (via USB output) and a plasma arc lighter. It combines energy storage, power conversion, and high-voltage generation in a single system.

3.2 COMPLETE CIRCUIT DIAGRAM



This circuit is divided into two main parts;

i. Power bank section;

- The 18650 lithium ion battery is connected to the TP4056 charging module which manages battery charging via a micro USB port
- The output of the TP4056 module is connected to a boost converter which steps up the voltage from 3.7V-5V suitable for USB powered devices
- A USB port is connected to the boost converter for charging phones and other devices.
- A slide or toggle switch is placed between the boost converter and USB output to control the power flow

ii. Plasma lighter section;

- A separate switch connects the battery directly to the plasma arc module allowing high voltage to be generated when needed

- Care is taken to isolate this high voltage part from the USB circuitry to avoid interference or damage

3.3 IMPLEMENTATION a) Battery preparation

- A good quality 18650 battery is selected and tested with a multimeter
- The battery is pot welded or soldered to the tp4056 charging module

b) Power bank configuration

- The output of the tp4056 module is connected to the input of the boost converter
- The 5v output from the boost converter is wired to a USB port and controlled via a switch

c) Plasma lighter integration

- A separate circuit is created where the battery powers the plasma lighter module through a dedicated switch
- Proper insulation is applied around high voltage path to ensure safety

d) Enclosure and finishing

- All components are securely arranged inside a custom or recycled case using shrink tubing or a 3d-printed housing
- Openings are made for the USB port, charging ports, switches and plasma arc tips

The entire device is tested thoroughly to ensure both functions (power bank and lighter)

3.4 SAFETY CONSIDERATION

- Overcharge and short circuit protection are handled by the TP4056 module
- The high voltage plasma arc is only activated when required and is isolated from other components.

CHAPTER FOUR

1.0 FINDINGS, DATA ANALYSIS, CONCLUSIONS AND RECOMMENDATIONS

4.1 FINDINGS

The DIY power bank with integrated plasmas lighter project led to several important findings during its development, assembly, and testing phases.

✓ Feasibility of integration.

- it is a technically feasible to integrate a plasma arc lighter with a portable power bank using compact and affordable components.
- Both functionalities (charging and ignition) can share the same battery source with the appropriate circuit protection.

✓ Power consumption

- The plasma lighter draws a significant current (typically 2-5A) during operation which can rapidly drain the battery if overused
- Efficient use and limited activation duration are crucial for preserving battery life of the power bank

✓ Battery selection and safety

- Lithium ion (18650) offer the best balance between capacity and size
- A battery management system is necessary to prevent overcharging, over discharging and short circuits

✓ Circuit design challenges

- Isolating the power draw of the plasma lighter from the USB output was necessary to avoid voltage drops affecting charging performance
- The inclusion of a boost converter and dual switch control helped manage power distribution effectively

✓ Compactness vs heat management

- Housing all components in a compact enclosure was possible but presented challenges in heat dissipation
- Proper ventilation or intermittent is required to avoid over heating

✓ User safety and durability

- Insulating and shielding high voltage areas of the plasma lighter was essential to prevent accidental electric shocks
- With robust casing and secure wiring, the device demonstrated handling under normal operation conditions

✓ Practical and usability

- The final prototype performed well in both functions-reliably charging devices and igniting materials
- Users appreciate the dual purpose functionality especially for outdoor or emergency use

4.2 DATA ANALYSIS

1.BATTERY PERFORMANCE

Parameter	value
Battery type	18650 Li-ion(3.7V-5V)
Number of cells	2 (in parallel or series with protection circuit)
Total capacity	5200mAh (parallel)
Output voltage(USB port)	5V (regulated via boost converter)
Charging time(USB input)	4.5 hours

Insight: The battery performed efficiently under normal power bank usage but showed rapid voltage drop when the plasma lighter was used repeatedly.

2.PLASMA LIGHTER PERFORMANCE

Parameter	Value
Arc value	~3.7V input n ~3000V at arc tip
Current Draw	2-5 A momentary burst
Arc Duration	5-7 seconds per activation (auto shutoff)

3.USB CHARGING OUTPUT ANALYSIS

Input current	Charging time	Battery level
~1.0A	~2.5	0% n 100%
~0.2A	~1 hour	10% n100%
~0.5A	Continuous	-

Insight; USB output was stable with regulated 5V. the boost converter maintained consistent voltage under light-to-light moderate loads.

4. THERMAL ANALYSIS

Component	Max Temp (°C)	Safe Range	Notes
Battery (under load)	42°C	<60°C	Within safe limits with airflow
Plasma lighter circuit	65°C	<80°C	Short use only recommended
Boost converter	38°C	<70°C	Efficient thermal performance

Insight; No thermal runaway was observed. Adequate spacing and occasional use of the lighter prevent overheating.

5.OVERALL EFFICIENCY

Function	Efficiency
USB Charging	~85% (with boost converter loss)
Plasma Lighter Arc	~60% (due to high voltage conversion loss)

Insight; The power bank mode was more energy-efficient than the lighter function.

Optimization can improve arc efficiency (e.g., through better drivers or energy storage capacitors)

4.3 PROJECT BUDGET

Component	Specification	Quantity	Price (Ksh)
18650 Li-ion Battery	3.5V,2600mAh	2	300
Flyback transformer	Boost type	1	1000
D313 transistor	3.7-5V ,2A output	1	100
Plasmas arc lighter module {stainless needles}	High voltage dual arc	2	50
Power bank charging module	Female USB-A and C	1	500
Push button switch,jumper wires,220k resistor,heatsink	Momentary type	1	600
PCB board , Enclosure and connecting wires.	PCB and Board enclosure	-	500
TOTAL			3050

4.4 CONCLUSION

This project successfully demonstrates the design and construction of a DIY power bank with a plasma lighter. By combining portable power storage with a rechargeable high-voltage

Take ignition system, the device offers practical dual functionality in a compact form.

The power bank components efficiently charge various USB powered gadgets while the plasma lighter provide reliable ignition without the need for fuel, making it ideal for outdoor and emergency use.

The project enhanced understanding of key electronics concepts such as battery charging, energy management and safe high voltage handling, it also encourages hands-on skill like soldering, circuit integration and enclosure designs.

4.5 RECOMMENDATION

To further improve the device and ensure better performance and safety, the following recommendations are made;

- Battery Expansion; use multiple 18650 cells in parallel to increase the total capacity and extend both charging and lighter usage time.
- Indicator LEDS; Add LED indicator to show charging status for both power bank and plasma lighter.
- Protective circuitry; Include fuses or additional protective circuit to safeguard against overcurrent or accidental short circuit.
- Separate power regulation; Consider using separate voltage regulators or a dual battery system to isolate the plasma lighter from the USB charging path effectively.

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4.7 APPENDICES

4.7.1 APPENDIX 1

User Warnings & Symbols

![High Voltage](https://img.icons8.com/ios/50/high-voltage.png) *HIGH VOLTAGE* - Do not touch electrodes.

![Flammable](https://img.icons8.com/ios/50/fire-hazard.png) *FLAMMABLE* - Keep away from gases/alcohol.

![Battery](https://img.icons8.com/ios/50/charged-battery.png) *NO DISASSEMBLY* - Risk of battery explosion.

APPENDIX 2; T00LS USED

- Soldering iron and wire
- Multimeter (for voltage and current testing)
- Screw drivers and pliers
- Hot glue gum for fixing components
- 3D printer or cutting tools (for custom casing)